

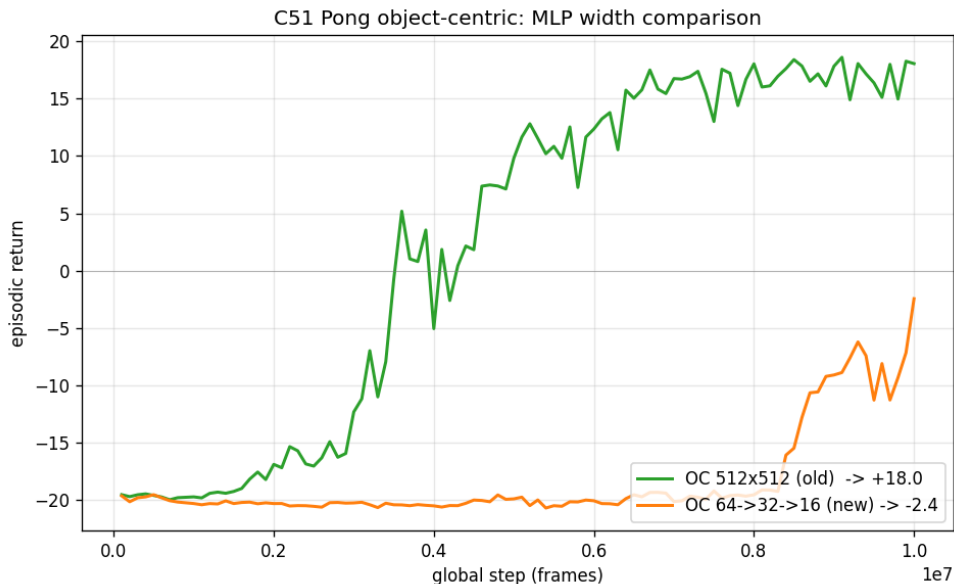
C51 on Pong (object-centric): MLP width comparison

Group 27 • Topic 27 • JAXAtari baselines • supplement to the C51/Pong report

Following the feedback that the 512×512 object-centric MLP was “too wide”, we re-ran object-centric (OC) C51 on Pong with a narrow tapered net ($64 \rightarrow 32 \rightarrow 16$). Everything else was held fixed (normalization, 10M frames, 8 envs, all hyperparameters). **The narrow net is much slower to converge:** it does eventually learn, but a 10M-frame budget cuts it off mid-climb.

OC network	Layers	Final return (10M)	Status at 10M
512×512 (old)	2	+18.0	solved, plateaued
$64 \rightarrow 32 \rightarrow 16$ (new)	3	-2.4	still rising steeply ($\approx +4.7$ / 100k)

What the new run does. It sits flat at ≈ -20 (losing every game) until ~ 8.3 M frames, then takes off sharply: $-19.2 \rightarrow -2.4$ over the last 1.7M frames and still accelerating at the end (not a plateau). Two side channels confirm this is genuine learning, not noise: episodic length jumps from ~ 900 to ~ 2700 (real rallies, not concessions) and the mean Q-value recovers from ≈ -1.4 back through 0 to $+0.29$, in lockstep with the return.



Object-centric learning curves overlaid. Old 512×512 (green) takes off at ~ 2 M and plateaus at $+18$ by ~ 6 M; new $64 \rightarrow 32 \rightarrow 16$ (orange) is flat until ~ 8.3 M, then a steep, unfinished climb to -2.4 at 10M.

Why it got worse. It is a convergence-*speed* gap, not a capability gap. The narrow net has far fewer parameters, so it needs many more gradient steps before the value function is accurate enough to produce a useful greedy policy (hence the ~ 8 M stuck phase). The tightest spot is the final 16-unit layer feeding the distributional head (action_dim $\times n_{\text{atoms}} = 6 \times 51 = 306$ outputs): a $16 \rightarrow 306$ bottleneck is severe (even CleanRL’s CartPole C51 uses $(120, 84)$). The new net is also deeper *and* narrower, which trains more slowly without residual connections.

Takeaway. Normalization—not width—was the real fix for the original OC collapse: even this tiny net learns. But width buys convergence speed within a fixed budget. We therefore recommend a middle ground (e.g. $128 \rightarrow 64 \rightarrow 32$, smaller than 512^2 but without the 16-unit bottleneck), or re-running $64 \rightarrow 32 \rightarrow 16$ at 15–20M frames purely to document the speed cost.